DO NOW — RECALL

Lesson 079 · Bubble Sort: Adjacent Swaps and Passes

Name: _____ Class: _____ Date: _____

1. Which elements does bubble sort compare?

2. What is placed by one ascending pass?

3. What does swap do?

4. Swap elements at indexes 0 and 1 in array [4,2,6]. Give the new array.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Adjacent pairs.

2. Model answer

Largest unsorted value at end.

3. Model answer

Exchanges two stored values.

4. Model answer

[2,4,6].

By the end, you can…

• Compare adjacent pairs, swapping them when out of ascending order.

• One completed ascending pass places the largest remaining element at the right end of the unsorted prefix.

• Straightforward bubble sort is $O(n^2)$ worst time and O(1) extra memory; an early-exit flag can give O(n) best time.

Vocabulary — know these terms

bubble sort — Compare/swap adjacent values.

pass — Sweep through neighbouring pairs.

swap — Exchange two elements.

invariant — Always true state after given stage.

NEW LEARNING

Read, annotate and explain

Understand

Compare adjacent pairs, swapping them when out of ascending order.

Trace

One completed ascending pass places the largest remaining element at the right end of the unsorted prefix.

Explain

Straightforward bubble sort is $O(n^2)$ worst time and $O(1)$ extra memory; an early-exit flag can give $O(n)$ best time.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Show full first pass [4,1,3,2].

Check the working

[1,4,3,2]→[1,3,4,2]→[1,3,2,4].

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Give all states from first pass [5,2,4,1]. [6]

Question 2

Give pass-end arrays for [5,2,4,1] until sorted. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain bubble-sort worst-time and extra-space complexity and benefit of early-stop. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

After ascending first bubble pass, smallest value must be fixed at index0.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.